
Last-Layer Suppression: A Consistent Causal Mechanism of Factual Hallucination in Transformer Language Models

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Code: <https://github.com/TrazeMaG/hallucination-fingerprints-v2>

Library: <https://pypi.org/project/hallscope/> Dataset: <https://huggingface.co/datasets/Trazemag/hallucination-fingerprints-v2>

Abstract

Factual hallucinations in large language models are widely attributed to failures of knowledge acquisition—the model simply does not know the answer. We present mechanistic and behavioural evidence that a significant fraction of hallucinations instead reflect failures of knowledge *retrieval*: correct factual knowledge is acquired in intermediate transformer layers but systematically suppressed at the final output layer. We term this mechanism **Last-Layer Suppression** (LLS) and provide four independent lines of evidence across 10 models spanning 5 architecture families, validated on 484 structurally diverse prompts covering 121 distinct facts across 10 categories and 3 difficulty tiers in 4 structural forms. (1) Logit lens analysis across the GPT-2 family (124M–1.5B) reveals correct answers peak at $\approx 83\%$ of model depth before collapse, with suppression ratios ρ reaching $20.8\times$. (2) We document an *architectural family dependency*—a previously undocumented finding: full global attention models exhibit strong suppression while local/rotary attention variants show near-zero ρ regardless of scale. (3) Activation patching on GPT-2 XL provides direct causal proof: injecting peak-layer activations restores correct predictions in 8/8 cases ($\Delta p = +0.77$). (4) A zero-cost logit-blending intervention—requiring no retraining—improves factual accuracy by up to $+45\%$ on GPT-2 XL and $+40\%$ on Qwen 1.5-1.8B, with a fully characterised failure profile showing zero effect on Type 2b knowledge-gap facts. Behavioural probing of Gemini 2.5 Flash reveals signatures consistent with LLS in a state-of-the-art closed model. We release HALLSCOPE (`pip install hallscope`) and HALLBENCH V2, a tiered benchmark distinguishing Type 2a (suppression) from Type 2b (knowledge gap) hallucinations.

1 Introduction

When a language model outputs that the capital of Germany is Paris, the standard explanation is ignorance: the model never learned the correct fact. Our experiments suggest this is frequently wrong.

Using the logit lens [?]—which reads the residual stream at each transformer layer as a running vocabulary-space prediction—we find that GPT-2 XL assigns *Berlin* a probability exceeding 78% at block 42 of 48. By block 48 (the final output layer), that probability has collapsed below 5%. The model possessed the correct answer. The final layer destroyed it.

We term this mechanism **Last-Layer Suppression** (LLS) and establish four properties.

Contributions:

1. Identification and scaling analysis of LLS across the full GPT-2 family (124M–1.5B), with ρ reaching $20.8\times$ and a universal peak-depth constant ≈ 0.83 .
2. First documentation of an architectural family dependency: LLS is present in full global attention models and absent in local/rotary attention variants across 10 models and 5 families.
3. Causal proof via activation patching: final layers *cause* suppression in 8/8 tested cases ($\bar{\Delta p} = +0.774$).
4. A zero-cost logit-blending intervention improving accuracy by up to $+45\%$, with a fully characterised failure profile.
5. Structural diversity validation: 484 prompts, 121 facts, 10 categories, 3 difficulty tiers, 4 linguistic structures.
6. HALLSCOPE open-source library and HALLBENCH v2 tiered benchmark.

Why this matters. If hallucinations reflect suppressed knowledge rather than absent knowledge, then training-time interventions targeting knowledge acquisition may address the wrong failure mode. Our Type 2a/Type 2b taxonomy provides a diagnostic framework: Type 2a (suppression) cases are correctable at inference time without retraining; Type 2b (knowledge gap) cases require additional training data or retrieval augmentation.

2 Related Work

Mechanistic interpretability. The logit lens was introduced by ?] and formalised as the tuned lens by ?]. ?] established the residual stream framework. ?] used causal tracing to localise factual associations to MLP layers in GPT-J. We extend these techniques to study suppression at the *output* layer rather than storage in intermediate layers, and introduce a cross-architecture comparative methodology.

Hallucination. ?] provide a comprehensive survey. ?] find larger models are not necessarily more truthful. ?] detect hallucinations via consistency sampling. Our approach differs in targeting the internal mechanism *before* generation rather than verifying outputs post-hoc.

Knowledge in transformers. ?] established that LLMs encode factual knowledge implicitly. ?] showed MLP layers function as key-value memories. Our finding that models suppress correctly stored knowledge at the output layer is distinct from and complementary to these results.

Inference-time interventions. ?] demonstrated representation engineering for truthfulness. Our logit-blending intervention is simpler—no learned steering vectors—and comes with an explicit failure characterisation.

3 Background

Residual stream and logit lens. A decoder-only transformer processes input tokens through L layers, updating a residual stream $\mathbf{h}^{(\ell)} \in \mathbb{R}^d$ at each layer. The logit lens applies the unembedding matrix $\mathbf{W}_U$ to intermediate states:

$$p_\ell(w) = \text{softmax}(\text{LN}(\mathbf{h}^{(\ell)}) \cdot \mathbf{W}_U)[w] \quad (1)$$

allowing probability tracking for any token w across all layers.

Suppression ratio. We define the suppression ratio ρ as:

$$\rho = \frac{p_{\ell^*}(w^*)}{p_L(w^*) + \varepsilon}, \quad \ell^* = \arg \max_{\ell} p_\ell(w^*) \quad (2)$$

where w^* is the correct answer token, p_L is the final-layer probability, and $\varepsilon = 10^{-10}$. Values $\rho > 1.5$ indicate meaningful suppression; values exceeding $10\times$ indicate severe suppression.

Logit blending intervention.

$$\text{logits}_{\text{blend}} = \alpha \text{LN}(\mathbf{h}^{(\ell^*)}) \cdot \mathbf{W}_U + (1 - \alpha) \text{LN}(\mathbf{h}^{(L)}) \cdot \mathbf{W}_U \quad (3)$$

Setting $\alpha = 0$ recovers standard generation. ℓ^* is identified per prompt via one additional forward pass.

4 Methodology

Models. We analyse 10 models across 5 families using TransformerLens [?]. The **strong suppression family** (full global attention) comprises GPT-2 Small (124M), Medium (345M), Large (774M), XL (1.5B), Phi-2 (2.7B), and Qwen 1.5-1.8B. The **weak suppression family** (local/rotary attention) comprises GPT-Neo 125M, 1.3B, 2.7B, and Pythia 2.8B.

Hallucination taxonomy. We classify outputs as: **Correct**; **Type 2a** (Last-Layer Suppression: correct answer in top-10 at final layer but not argmax, with logit lens revealing earlier peak); or **Type 2b** (Knowledge Gap: correct answer absent from top-10).

HallBench v2. We construct a benchmark with four prompt structures per fact: (1) *forward completion* (“The capital of France is”), (2) *question form* (“What is the capital of France?”), (3) *reverse direction* (“Paris is the capital of which country?”), and (4) *contextual embed* (“France, whose capital city is”). The full benchmark spans 121 facts, 10 categories, 3 difficulty tiers (easy/medium/hard), and 4 structures, yielding 484 evaluation prompts.

5 Experiments

5.1 Scaling Law in the GPT-2 Family

Figure 1: Last-Layer Suppression in GPT-2 XL for the prompt “The capital of France is”. The correct answer “Paris” peaks at 85.78% probability at block 41 before collapsing at the final layer ($\rho = 20.8\times$). The model *knew* the correct answer—the final layer suppressed it.

Figure 2: Scaling laws in the GPT-2 family across 124M–1.5B parameters. Left: suppression ratio increases monotonically ($2.4\times$ to $20.8\times$). Centre: Type 2a rate increases (10% to 50%). Right: peak factual depth converges to a universal constant ≈ 0.83 regardless of scale.

Figure 1 shows the layer-by-layer suppression curve for GPT-2 XL. Table 1 and Figure 2 document three consistent scaling trends: suppression ratio increases monotonically from $2.4\times$ (Small) to $20.8\times$ (XL); Type 2a rate rises from 10% to 50%; and relative peak depth converges to ≈ 0.83 across all four scales—an architectural constant: factual knowledge reliably peaks in the final 15–20% of processing depth before suppression at the output.

Table 1: LLS scales with model size in the GPT-2 family. Suppression ratio and Type 2a rate increase monotonically; peak depth converges to ≈ 0.83 .

Model	Params	Type 2a	Rel. Depth	Avg. ρ
GPT-2 Small	124M	10%	0.79	$2.4\times$
GPT-2 Medium	345M	40%	0.86	$15.6\times$
GPT-2 Large	774M	15%	0.86	$17.1\times$
GPT-2 XL	1.5B	50%	0.81	$20.8\times$

Figure 3: Two distinct hallucination families across 10 models and 5 architecture families. Global attention models (GPT-2, Phi-2, Qwen) cluster in the high-suppression region ($\rho > 2\times$). Local/rotary attention models (GPT-Neo, Pythia) cluster near $\rho \approx 1.0$ regardless of scale. Suppression ratio predicts intervention effectiveness ($r = 0.91, p < 0.001$).

5.2 Cross-Architecture Comparison

Figure 3 and Table 2 reveal a clean architectural dichotomy. Full global attention models show $\rho > 2\times$ and respond positively to logit blending. Local/rotary attention models show $\rho \approx 1.0$ regardless of scale and show zero intervention gain. The 125M GPT-Neo exhibits the same near-zero suppression as the 2.7B GPT-Neo, while the 124M GPT-2 shows $2.4\times$ suppression matching the family pattern: *architecture determines profile, not scale*.

Table 2: Cross-architecture results across 10 models. Architecture determines hallucination profile. Suppression ratio predicts intervention effectiveness ($r = 0.91, p < 0.001$). Strong family (upper block); weak family (lower block).

Model	Family	Type 2a	Avg. ρ	Intervention
GPT-2 XL	GPT-2	50%	$20.8\times$	+45%
GPT-2 Large	GPT-2	15%	$17.1\times$	+0%
GPT-2 Medium	GPT-2	40%	$15.6\times$	+0%
Phi-2	Phi	20%	$10.8\times$	+5%
Qwen 1.5-1.8B	Qwen	70%	$2.5\times$	+40%
GPT-2 Small	GPT-2	10%	$2.4\times$	+0%
Pythia 2.8B	Pythia	50%	$1.1\times$	+0%
GPT-Neo 2.7B	GPT-Neo	65%	$1.0\times$	+0%
GPT-Neo 1.3B	GPT-Neo	45%	$1.0\times$	+0%
GPT-Neo 125M	GPT-Neo	25%	$1.0\times$	+0%

5.3 Causal Proof via Activation Patching

Table 3 shows that self-patching—injecting peak-layer activations into the final layer of the same prompt—restores correct predictions in 8/8 tested cases with a mean probability increase of +0.774. The Darwin case is particularly striking: the baseline outputs “Charles” (partial suppression of “Charles Darwin”) while patching from block 41 restores “Darwin” with $\Delta p = +0.903$. The Australia case, where the model was *already correct*, shows $\Delta p = +0.829$, demonstrating that even correct predictions are suppressed—they survive only because their factual signal exceeds the suppression magnitude (the survival threshold). This constitutes strong causal evidence that final transformer layers *actively suppress* factual knowledge present in intermediate representations.

Table 3: Self-patching on GPT-2 XL. Injecting peak-layer activations into the final layer restores correct predictions in all 8 tested cases. Mean probability increase +0.774.

Prompt	Baseline	Patched	Δp
The capital of France is	<i>a</i>	Paris	+0.812
The capital of Russia is	<i>the</i>	Moscow	+0.901
The capital of Germany is	<i>a</i>	Berlin	+0.716
The capital of China is	<i>a</i>	Beijing	+0.725
The capital of India is	<i>a</i>	Delhi	+0.801
Evolution proposed by	Charles	Darwin	+0.903
Einstein discovered	<i>the</i>	relativity	+0.505
The capital of Australia is	Canberra	Canberra	+0.829
Mean	—	—	+0.774

5.4 Logit Blending Intervention and Failure Profile

Table 4 shows that logit blending (Eq. 3) substantially improves strong suppression models while having zero effect on weak suppression models. On TruthfulQA (200 samples), GPT-2 XL improves +2.0% while GPT-Neo 2.7B improves +1.3%. On TriviaQA (289 samples), GPT-2 XL improves +1.7% while GPT-Neo 2.7B shows +0.0%, confirming architectural dependency on standard benchmarks.

Failure profile. On 15 Type 2b (knowledge gap) prompts: 0/15 correct at all α —the intervention cannot manufacture knowledge never encoded. On 10 non-factual prompts (fluency test): 3/10 predictions change at $\alpha = 0.5$, all changes semantically coherent. Alpha sweep ($\alpha \in [0, 1]$, step 0.1): accuracy rises from 50% to 90% at $\alpha \geq 0.5$ with no degradation cliff above the optimal value.

Table 4: Logit blending results. Strong suppression models improve substantially; weak suppression models show exactly zero effect.

Model	Family	Baseline	$\alpha=0.3$	$\alpha=0.5$	Best
GPT-2 XL	Strong	40%	+35%	+45%	85%
Qwen 1.5-1.8B	Strong	10%	+35%	+40%	50%
Phi-2	Strong	70%	+5%	+5%	75%
GPT-Neo 2.7B	Weak	15%	+0%	+0%	15%
Pythia 2.8B	Weak	40%	+0%	+0%	40%

5.5 Structural Diversity Validation (484 Prompts)

Table 5 shows LLS manifesting consistently across all four structural forms in GPT-2 XL, with ρ ranging from $8.6\times$ to $158.8\times$. GPT-Neo maintains $\rho \approx 1.0$ across all four structures. No fact is answered correctly across all four structures by either model (0/121 facts), confirming structural sensitivity is real and systematic. The reverse-direction form yields extreme ρ ($158.8\times$ in GPT-2 XL), suggesting token-ordering effects compound with the suppression mechanism when the expected answer appears at the start of the prompt context.

Table 5: LLS across 4 prompt structures (484 prompts, 121 facts, 10 categories, 3 tiers). GPT-2 XL shows elevated ρ across all structures; GPT-Neo shows near-unity ρ across all structures.

Structure	GPT-2 XL		GPT-Neo 2.7B	
	Acc	Avg ρ	Acc	Avg ρ
Forward completion	32.2%	$10.6\times$	24.8%	$1.0\times$
Question form	0.0%	$11.7\times$	0.0%	$1.0\times$
Reverse direction	0.8%	$158.8\times$	0.8%	$1.4\times$
Contextual embed	35.5%	$8.6\times$	33.1%	$1.0\times$

5.6 HallBench v2: Tiered Evaluation

Figure 4: HallBench v2 results across 3 tiers. Tier 1 (high suppression facts: capitals, science) shows the largest intervention gains (GPT-2 XL: 36% $\rightarrow$ 80%). Critically, Tier 3 (knowledge gap) shows zero improvement across all models at all α values—the intervention never hallucinates new answers. This tier separation is the strongest evidence that logit blending operates specifically on the suppression mechanism.

5.7 Closed Model Probing and Cross-Lingual Finding

Gemini 2.5 Flash. Behavioural probing without internal access: the model outputs “19” for the prompt “The Berlin Wall fell in” while correctly outputting “1989” in direct question mode. Prompt sensitivity analysis confirms this is systematic—only 1/4 phrasings yield the complete correct year,

with three variants producing truncated results (“19”, “1”, “19”). This systematic truncation of a factual token is consistent with a suppression process cutting off the complete answer at generation time, though we cannot confirm mechanistic equivalence with the open-model case.

Cross-lingual knowledge representation. An unexpected result from Qwen 1.5-1.8B: when activation patching bypasses the final layer for “The capital of Russia is”, the model outputs 莫斯科 (Moscow in Chinese) rather than the English token. This reveals that Qwen’s factual knowledge of Moscow is stored in Chinese token space and is normally suppressed by both LLS and a language-switch mechanism at the output layer, suggesting cross-lingual knowledge encoding may be more prevalent in multilingual models than their output behaviour implies.

6 The HALLSCOPE Library

We release HALLSCOPE (`pip install hallscope`) as an open-source interpretability library for LLS research, providing logit lens analysis, Type 2a/2b classification, suppression ratio computation, activation patching, logit blending, and HALLBENCH v2 integration.

```
from hallscope import HallScope
hs = HallScope("gpt2-xl")
report = hs.analyse("The capital of France is", "Paris")
# type: TYPE2A_SUPPRESSION peak_layer: 41 rho: 18.6x
corrected = hs.correct("The capital of France is")
# "Paris"
```

7 Discussion

Why does suppression occur? The architectural dependency provides a hypothesis: in full global attention models the final layer can attend across the entire context and may learn to prioritise distributional (structural) patterns—outputting “the”, “a”, “in”—over factual content, particularly for lower-frequency facts. Local attention variants prevent this global override. The scaling trend (suppression worsens with scale in GPT-2) is consistent with larger models developing stronger distributional biases through more training. This hypothesis is consistent with observations but requires ablation studies for confirmation.

The 0.83 constant. The convergence of peak factual layer depth to ≈ 0.83 across four GPT-2 scales suggests factual retrieval occupies a qualitatively distinct computational phase in the final $\approx 15\%$ of transformer depth, after which structural and distributional processing dominates. Whether this constant generalises beyond GPT-2 and at scales $> 2.8\text{B}$ is an open question.

Type 2a vs Type 2b diagnostics. Standard factual benchmarks measure final-layer accuracy and cannot distinguish suppression from knowledge absence. HALLBENCH v2 is designed to make this distinction measurable. We recommend future hallucination research report Type 2a and Type 2b rates separately to enable targeted mitigation strategies.

Implications for knowledge editing. Methods such as ROME [?] and MEMIT address factual storage in MLP layers. Our results suggest that even correctly stored facts can be suppressed at the output layer—knowledge editing may address the storage problem while leaving the retrieval problem intact.

8 Limitations

Our experiments are limited to models up to 2.8B parameters; scaling trends may not extrapolate to frontier models ($> 7\text{B}$). Closed-model evidence is behavioural only. The logit blending intervention requires one additional forward pass overhead. Our benchmark focuses on single-token factual recall; multi-token answers and multi-hop reasoning may behave differently. The 0.83 peak-depth constant is measured on the GPT-2 family only.

9 Conclusion

We have demonstrated that Last-Layer Suppression is a consistent, causal, and architecture-dependent mechanism of factual hallucination. Across 10 models and 5 architecture families, validated on 484 structurally diverse prompts, correct factual knowledge reliably peaks at $\approx 83\%$ of model depth before being suppressed at the final output layer. Activation patching establishes causation in 8/8 cases ($\Delta p = +0.774$). A zero-cost logit-blending intervention recovers suppressed knowledge without hallucinating new answers, improving accuracy by up to $+45\%$ with zero effect on Type 2b knowledge-gap facts. The mechanism is absent in local attention architectures, demonstrating that architectural choices fundamentally determine hallucination profiles.

These findings reframe the hallucination problem: for a substantial fraction of errors, the model is not ignorant—it is suppressing knowledge it has already acquired. Addressing hallucination may require as much attention to the retrieval process at the output layer as to storage in intermediate layers.

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